SMART TASK SCHEDULER – PROJECT REPORT

Abstract

The Smart Task Scheduler is a Java-based software system designed to help users effectively plan, prioritize, and manage daily tasks. The system leverages priority-based logic, automated scheduling, task conflict detection, and deadline reminders to improve productivity and help users stay organized.

1. Introduction

Time management plays a crucial role in enhancing productivity, especially for students and working professionals. The Smart Task Scheduler provides a structured approach to managing tasks by allowing users to add, update, organize, and prioritize tasks. The system automates scheduling based on urgency and importance, making it a useful tool for efficient workload distribution.

2. Project Objectives

Provide a simple interface for adding and managing tasks.Implement a priority-based scheduling algorithm.Detect task conflicts and notify users.Generate reminders for upcoming deadlines.Enhance productivity through automated task organization.

3. Scope of the System

The Smart Task Scheduler focuses on lightweight personal task management. It does not integrate external calendars or cloud sync but provides a robust offline scheduling mechanism. The system is ideal for personal use, academic planning, or small team task tracking.

4. System Requirements

Software Requirements: Java JDK 8 or laterIntelliJ IDEA / Eclipse / VS Code (any IDE)**Hardware Requirements:**Minimum 4GB RAMAny standard Windows/Linux/Mac system

5. System Architecture

The system follows a modular design using Core Java OOP principles. Major components include:**Task Module:** Handles task creation, editing, and deletion.**Scheduler Module:** Applies priority-based logic to organize tasks.**Conflict Manager:** Detects overlapping or duplicate tasks.**Reminder Engine:** Generates alerts for nearing deadlines.**User Interface Module:** Provides console-based interaction.

6. System Workflow

1. User adds task details such as title, description, priority, and due date.
2. Scheduler arranges the tasks based on priority score.
3. Conflict detection ensures no overlapping tasks exist.
4. Reminders are generated for critical tasks.
5. Users view or update the task list as needed.

7. Key Features

✔ Priority-based automatic scheduling✔ Deadline reminders✔ Task conflict detection✔ Task categorization✔ Clean and modular Java architecture

8. Implementation Details

The application is developed using Java, utilizing classes, inheritance, and collections. ArrayList and HashMap are used to store tasks dynamically. The main logic resides in the Scheduler class, where priority scoring and conflict checks are executed.

9. Code Structure

com.app ■ ■■■ model ■ ■■■ Task.java ■ ■■■ service ■ ■■■ TaskManager.java ■ ■■■ Scheduler.java ■ ■■■ MainApp.java

10. Use Case Scenarios

Use Case 1: Student Assignment Planning
Students can schedule tasks based on deadlines and importance.

Use Case 2: Daily Routine Management
Users can plan daily goals and ensure timely completion.

11. Testing

The system underwent functional testing to validate task creation, scheduling accuracy, and conflict detection. All modules performed as expected.

12. Conclusion

The Smart Task Scheduler project successfully demonstrates a reliable and useful personal productivity tool built with Java. Its modular structure, automated scheduling logic, and task conflict system make it suitable for students and professionals looking to improve time management.

13. Future Enhancements

Integration with Google Calendar / OutlookMobile application versionCloud backup & syncGraphical user interface (GUI)